



Dying Red giant star

Very Large Telescope has captured planetary nebula of a dead giant star that has thrown off its outer layers, <https://digit.in/feb19-55>



Meteorite collision sparks Moon

A Meteorite Crashed Into the Moon during the Lunar Eclipse which could be spotted as a brief flash of light. <https://digit.in/feb19-56>

The first photo from the surface of the far side of the Moon

was a rover for in-situ exploration around the landing site. Communications to the far side is blocked by the Moon itself, so the Chang'e 4 mission was executed through a relay satellite launched in May 2018.

THE MAGPIE BRIDGE

The story of the weaver girl and the cowherd is one of the folk tales of China. This is the story of two lovers, who represent the stars Vega (the weaver girl) and Altair (the cowherd), who are banished to opposite sides of a river (the Milky Way galaxy). Once a year, a bridge of magpies, facilitates their meeting. The Chinese National Space Administration (CNSA) deployed a satellite called Queqiao, which means "the magpie bridge", in orbit around the L2 point of the Earth-Moon system, in what is known as a "halo orbit". As seen from the Earth, the orbit of the satellite looks like a halo around the Moon. The gravitational forces of the Earth and the Moon together work to keep satellites deployed at this point in place, which makes maintaining the orbit economical in terms of fuel consumption. Queqiao is the first satellite to operate from that orbit, and the location gives it an advantage. However, the very slight perturbations, including gravitational influences of the Sun and other planets, can disturb the orbit of the satellite. Which is why, the orbit of the satellite has to be maintained once every seven days. The orbital maintenance requires controlling the angle, altitude and speed of the satellite with high precision. Queqiao has an orbital period of 14 days. It can communicate with spacecraft on the far side of the Moon, and maintain a line of sight to the surface of the Earth. The main antenna for communications has a diameter of 4.2 meters, and is shaped like an umbrella. The satellite is designed for a mission duration of 3 years.

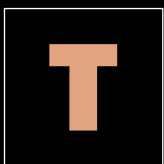
CHANG'E-4 PROBE

The Ranger 4 mission was actually the first spacecraft to land on the far side of the Moon. It was an impact

EXPLORING THE DARK SIDE OF THE MOON

The Chang'e 4 mission, named after the Chinese Moon goddess, is the first to execute a soft landing on the far side of the Moon, and explore the surface.

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he Moon is tidally locked to the Earth, and any observer on Earth can only see one side. The far side

is called the dark side not because it receives no sunlight, but because it has not been studied extensively by scientists on Earth, and little is

understood about it. The composition of the far side is expected to be different from the near side, if the Moon was formed in an impact event between the Earth and a Mars sized planet, soon after the formation of the Solar System. The crust is also suspected to be thicker on the far side, and can only be confirmed through in-situ examination. On January 3, 2019, the Chang'e 4 mission from China touched down on this uncharted territory. On board